

URL Shortener — Known Issues and Bugs

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: /home/dominic/Documents/fastapiurl

Report #: 18/20

1. **No rate limiting** — anyone can spam the shortener
 2. **No auth** — anyone can delete any URL
 3. **SQLite concurrency** — not safe for concurrent writes
 4. **Short codes** are not cryptographically random
 5. **No click analytics** beyond simple counter
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Known Issues & Bugs

Current Issues

- No automated tests for frontend
- SQLite not suitable for production concurrency
- No input sanitization for user-generated HTML
- WebSocket reconnection not implemented
- Error handling incomplete for edge cases

Limitations

Limitation	Impact	Workaround
SQLite	Concurrent writes bottleneck	Migrate to PostgreSQL
No caching	Repeated DB queries	Add Redis
No monitoring	No uptime tracking	Add health check aggregator
Manual deploy	No CI/CD	Add GitHub Actions

Technical Debt

- Inline CSS/JS in AI Chat Proxy (hard to maintain)
- No TypeScript (vanilla JS)
- No database migrations (auto-create)
- Mixed Python 3.10/3.11 features
- No environment validation at startup

Browser Compatibility

Browser	Status	Notes
Chrome 120+		Full support
Firefox 120+		Full support
Safari 17+		Minor CSS differences
Edge 120+		Full support

Performance Degradation Scenarios

1. Large PDF upload in RAG module (>10MB) may timeout
2. Multiple concurrent WebSocket connections increase memory
3. SQLite write contention under high load
4. OpenAI API latency affects chat response time

Security Known Issues

1. Rate limiting not implemented on all endpoints
2. No CSRF protection for state-changing requests
3. Security headers not configured
4. No audit log for admin actions
5. API keys in environment could be exposed via debug endpoints

Deprecated Features

- None currently identified